# Software Design Description 300

## ESP32-S3 Receiver – Overview

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## Overview

The ESP32-S3 is the receiver end of the trail camera system. It receives compressed WHT coefficients from the DevKitV1 over LoRa, reconstructs frames using inverse WHT and Gaussian smoothing against stored reference frames, and serves the reconstructed images to a mobile app via WiFi AP and WebSocket.

**Hardware:** ESP32-S3 WROOM | 8 MB PSRAM | UART to LoRa (RYLR998) | WiFi AP | No SD card

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## Module Hierarchy

```
main.c                       Entry point, init, task creation, heap monitor
|---- lora_uart.c            UART interface to RYLR998 LoRa module
|---- lora_bench.c           LoRa bench test echo slave (TEST_MODE_LORA_BENCH)
|---- image_reconstruct.c   Gaussian reconstruction from WHT coefficients
|---- wifi_ap.c              WiFi Access Point ("TrailCamera")
\---- ws_server.c            WebSocket server - streams frames to Flutter app
```

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## Operating Modes

### Production Mode (default)

Two functional domains, pinned to separate cores:

| Domain | Core | Tasks | Purpose |
|---|---|---|---|
| LoRa + Reconstruction | 0 | `lora_rx` | Receive packets, reconstruct frames |
| WiFi + WebSocket | 1 | httpd (async) | Serve frames to mobile clients |

### LoRa Bench Mode (`#define TEST_MODE_LORA_BENCH`)

Only the `lora_bench` echo slave task runs. WiFi and WebSocket are disabled. The S3 echoes received packets back to the DevKitV1 for RF link validation. Includes ready/start handshake protocol. See SDD004.

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## LoRa Interface

| Parameter | Value |
|---|---|
| UART | UART_NUM_2 |
| TX pin | GPIO 17 (ESP32 TX → module RX) |
| RX pin | GPIO 18 (module TX → ESP32 RX) |
| Baud | 115200 |
| Module address | 1 (receiver) |
| Network ID | 6 (must match DevKitV1) |
| Max payload | 240 bytes (RYLR998 limit) |

**Critical wiring note:** Swapping TX/RX on the S3 (GPIO17/GPIO18) silently kills all comms with no error output. See SDD004 Bug 1 context.

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## Image Reconstruction

### Reference Frame Storage

- 3 pan/tilt positions stored in PSRAM
- 3 x 307,200 bytes (640x480 grayscale) = ~921 KB
- Allocated with `heap_caps_calloc(MALLOC_CAP_SPIRAM)` at init

### Reconstruction Pipeline

Maps to `sim/codec.py:decode_frame()`:

1. **Parse packet header** — extract position index, scale, bounding box
2. **Identify active blocks** — blocks with nonzero WHT coefficients
3. **Dequantize** — multiply quantized values by `q_step`
4. **Inverse 2D WHT** — same butterfly as forward transform, divide by 64
5. **Gaussian smoothing** — smooth reconstructed diff to reduce block artifacts
6. **Add to reference** — `frame_out = reference[position] + smoothed_diff`
7. **Clamp** — pixel values to [0, 255]

### Frame Constants

| Parameter | Value | Notes |
|---|---|---|
| `FRAME_W` | 640 | Pixels per row |
| `FRAME_H` | 480 | Rows per frame |
| `FRAME_BYTES` | 307,200 | Grayscale |
| `BLOCK_SIZE` | 8 | WHT block dimension |
| `BLOCKS_W` | 80 | Blocks per row |
| `BLOCKS_H` | 60 | Block rows per frame |
| `NUM_POSITIONS` | 3 | Pan/tilt positions |

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## WiFi Access Point

| Parameter | Value |
|---|---|
| SSID | `TrailCamera` |
| Password | `wildlife1` (WPA2) |
| Channel | 6 |
| Max connections | 4 |
| IP | `192.168.4.1` |

- Created on boot using ESP-IDF WiFi AP API
- NVS flash initialized for WiFi state
- Event handler logs client connect/disconnect with MAC address

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## WebSocket Server

| Parameter | Value |
|---|---|
| Port | 80 |
| Path | `/stream` |
| Max clients | 4 |

| Parameter | Value |
|---|---|
| Protocol | Binary WebSocket |

**Frame format (binary message):**

| Byte offset | Size | Content |
|---|---|---|
| 0 | 1 | Position index (0=left, 1=center, 2=right) |
| 1-4 | 4 | Frame sequence number (uint32) |
| 5+ | FRAME_BYTES | Raw grayscale pixels, row-major |

Push-based: `ws_broadcast()` sends to all connected clients when a new frame is reconstructed. Disconnected clients are cleaned up on send failure.

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## Source Files

| File | Purpose | Status |
|---|---|---|
| `src/main.c` | Init, mode selection, task creation | Done |
| `src/lora_uart.c` | LoRa UART AT command interface | Done |
| `src/lora_bench.c` | Bench test echo slave + handshake | Done |
| `src/image_reconstruct.c` | Gaussian reconstruction from coefficients | Done (needs WHT coefficient parsing update) |
| `src/wifi_ap.c` | WiFi AP creation | Done |
| `src/ws_server.c` | WebSocket frame broadcast | Done |

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## Known Issues

1. **LoRa bench mode still enabled** — `TEST_MODE_LORA_BENCH` is `#define`'d in `config.h`. Must be commented out to enable production mode (WiFi + reconstruction).
2. **Power rail sag at high TX power** — S3 dev board LDO cannot supply RYLR998 at CRFOP=22. Currently limited to CRFOP=7. Needs bulk capacitor for field deployment (SDD004, Bug 5).
3. **Frame constants mismatch** — `config.h` defines 640x480 but WHT pipeline operates on 320x240. Needs alignment when FPGA pipeline is integrated.

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## References

- SDD000 — Development Standards
- SDD003 — LoRa Link Budget
- SDD004 — LoRa Bench Test & Debug
- SDD200 — ESP32-DevKitV1
- SDD500 — Simulation Pipeline
- `sim/codec.py:decode_frame()` — Reference reconstruction implementation